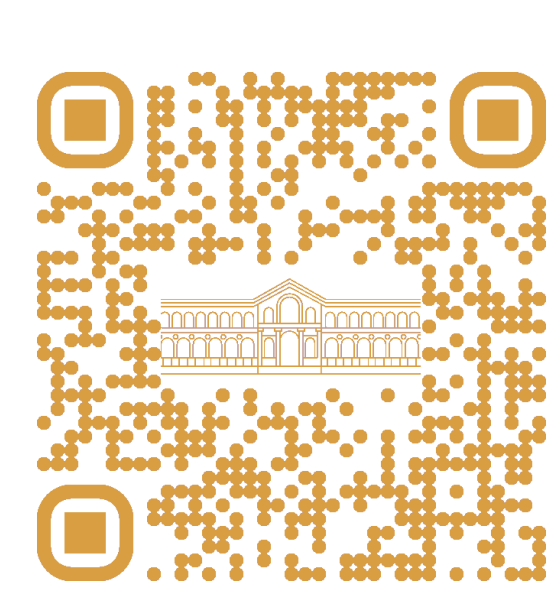




Instance-Guided Unsupervised Domain Adaptation for Robotic Semantic Segmentation



ICRA

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Context

Domain shift problem in semantic segmentation for indoor robotics

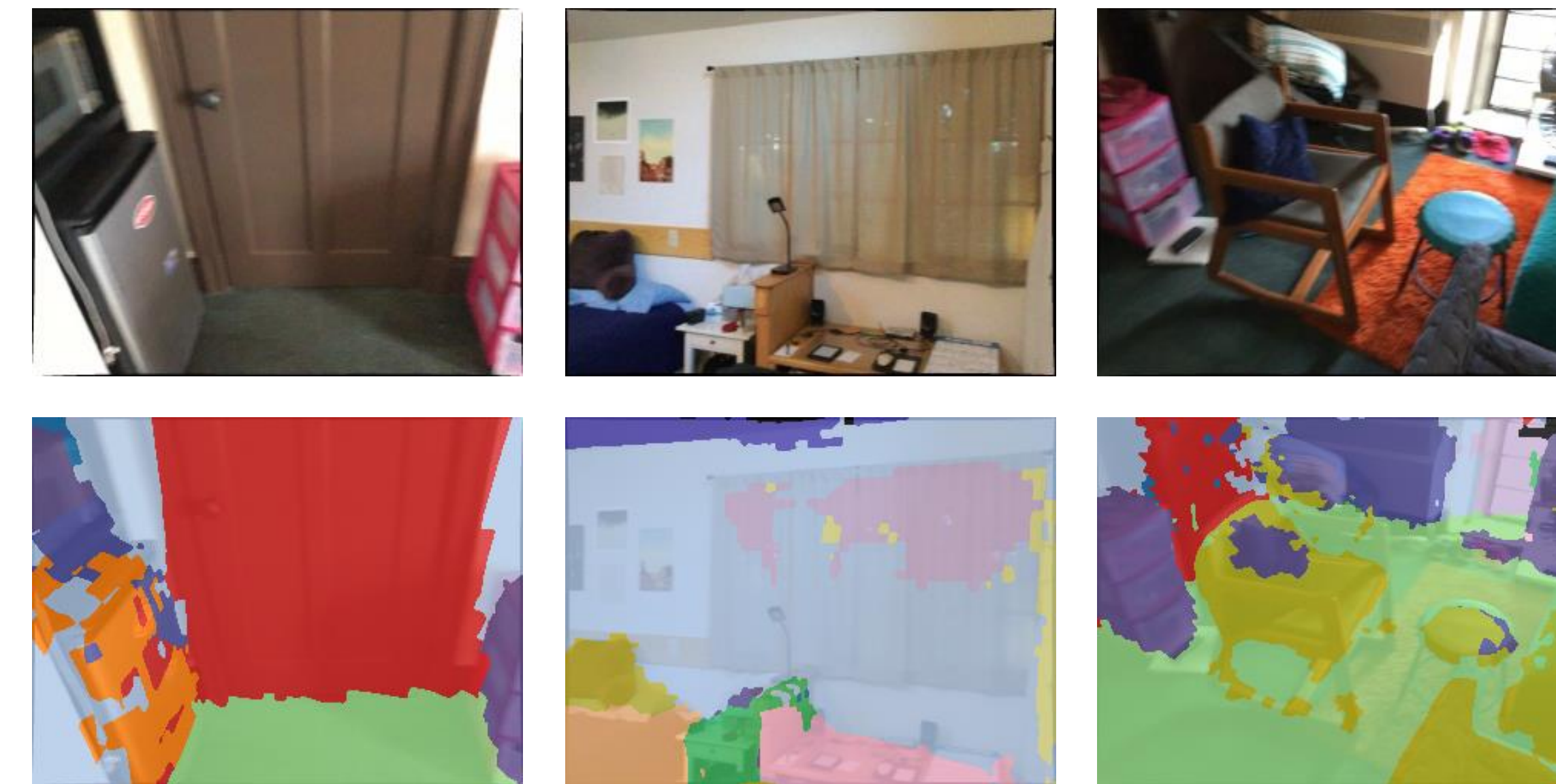
- Training on a **source domain**: general-purpose datasets, often in controlled conditions
 - Testing on a **target domain**: unseen environments, often with novel visual aspects
 - Fine-tuning with manual annotations is not practical in real-world robot deployments
 - Self-supervision with pseudo-labels is a common solution for unsupervised adaptation^[1]
- Performance degradation



Motivation

Multi-view consistent self-supervision

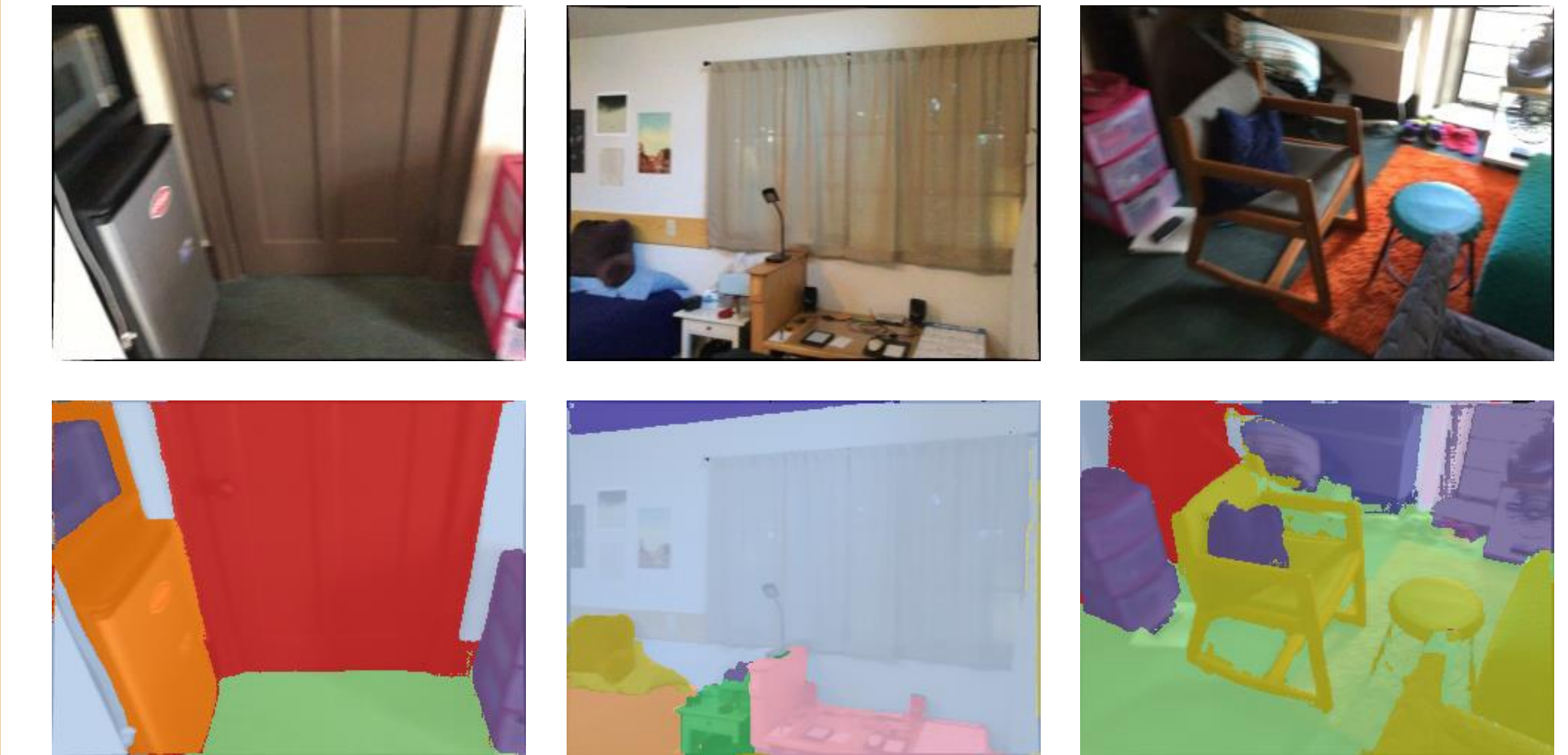
- Aggregating pseudo-labels from different point of views into the same 3D region (voxels in a 3D map) to force spatial consistency
- Obtaining spatially consistent pseudo-labels from the map
- **Limitations:**
 - Visual artifacts produced by the rendering process
 - Multiple categories can be assigned to the same object



Solution

Instance-guided unsupervised adaptation

- Revise pseudo-labels exploiting object-instance consistency: propagate pseudo-labels using guidance from an instance map
- We exploit the zero-shot instance segmentation model (Segment Anything v2^[3]) devising two complementary prompting strategies



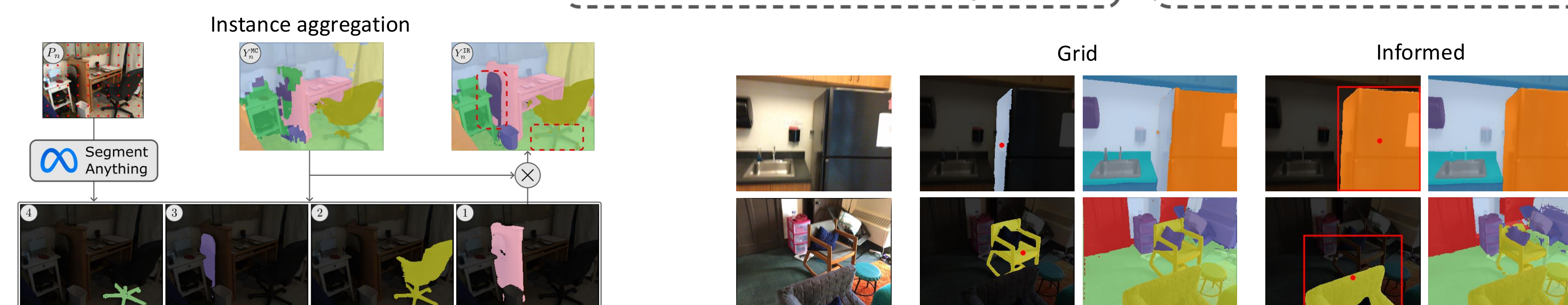
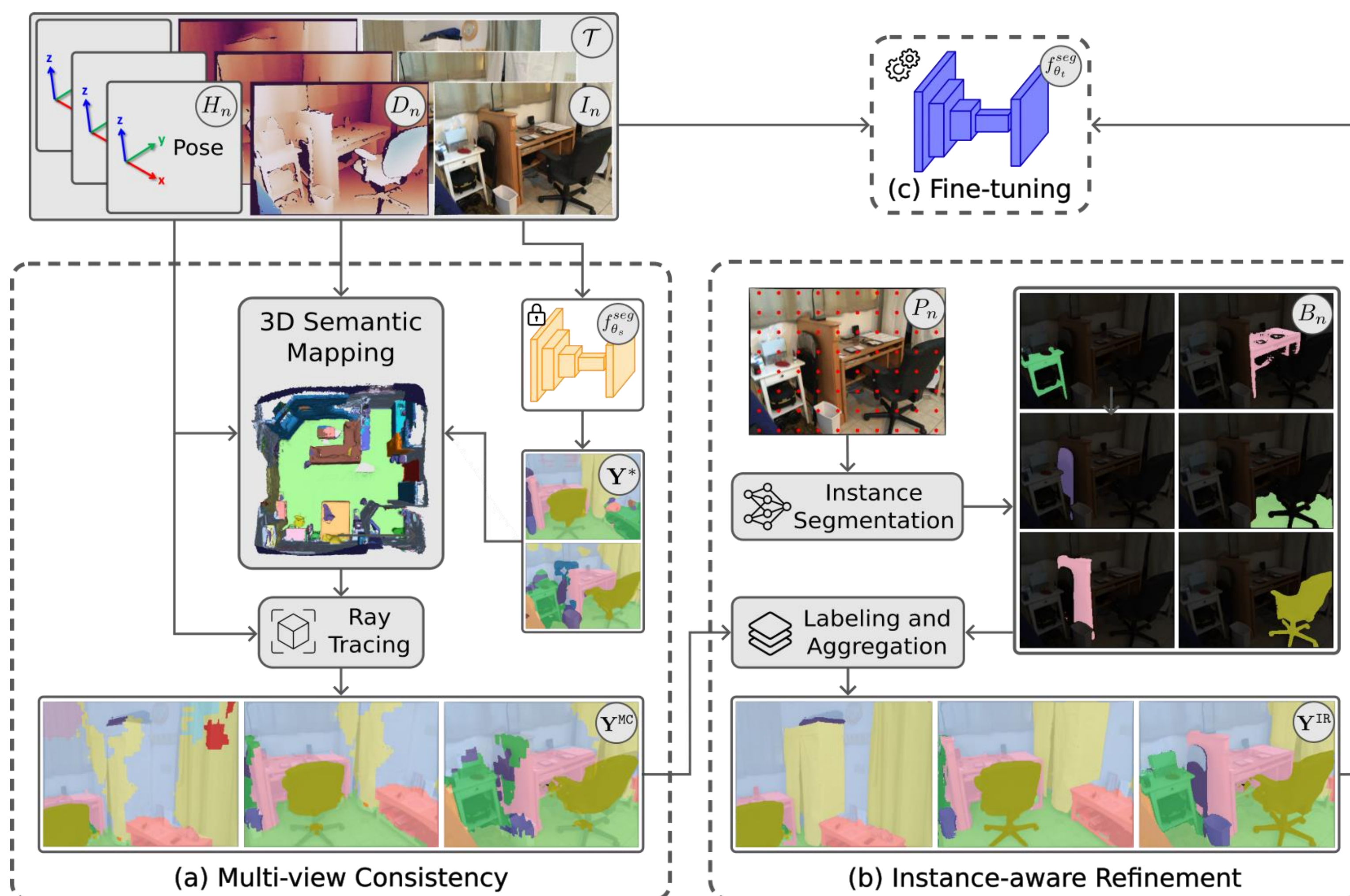
Method

a) Multi-view consistency

- Spatially-consistent pseudo-labels $\mathbf{Y}^{MC}$ are ray-traced from the map

b) Multi-view consistency

- Object instances B are obtained using SAM v2
 - **Grid**: the prompts are a grid of points uniformly distributed in the image
 - **Informed**: point and bounding box pairs derived from the pseudo-labels
 - **Combined**: pseudo-labels obtained with both strategies are used during training
- Object instances are labeled using the object categories in $\mathbf{Y}_n^{MC}$
- Labeled instances are aggregated in an instance-refined pseudo-label $\mathbf{Y}_n^{IR}$



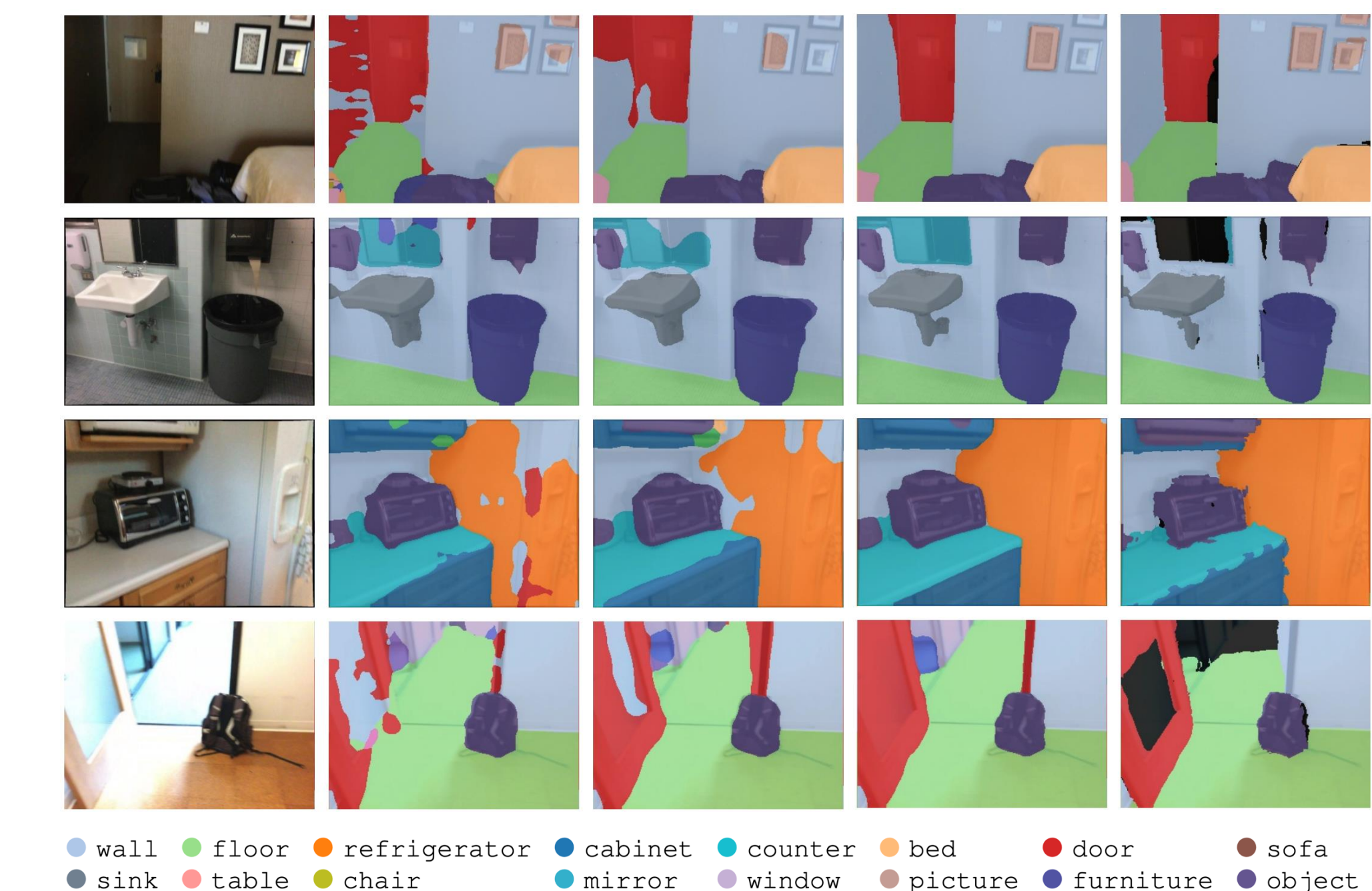
Results

Evaluation protocol

- **Dataset**: ScanNet^[4], environments 1-10 for adaptation/testing, 11-707 for initializing the source model
- **Model**: DeepLab v3
- **Mapping framework**: Kimera semantics
- **Baseline**: multi-view consistency^[2] (MC)

Results

- Our instance refinement (IR) step improves pseudo-labels quality
- Fine-tuning with our improved pseudo-labels improve model performance
- Prompting strategies performs similarly but Combined wins



Env	Source	Baseline		Our method		Our method		Our method	
		MC	Δ_{PT}	IR _G	Δ_{MC}	IR _I	Δ_{MC}	IR _{GI}	Δ_{MC}
1	47,1	49,6	5,3%	55	10,9%	51,2	3,2%	<u>54,3</u>	9,5%
2	44,1	35,2	-20,2%	39,6	12,5%	<u>39,6</u>	12,5%	39,3	11,6%
3	30,6	30,5	-0,3%	32,8	7,5%	34,4	12,8%	34,7	13,8%
4	55,3	55,8	0,9%	<u>57,2</u>	2,5%	56,2	0,7%	57,7	3,4%
6	49,2	50,5	2,6%	52,8	4,6%	54,7	8,3%	<u>53</u>	5,0%
7	52	54,4	4,6%	55,5	2%	57,2	5,1%	<u>56,2</u>	3,3%
10	57,5	63,2	9,9%	<u>68,7</u>	8,7%	67,6	7,0%	68,7	8,7%
Avg	48	48,5	0,4%	<u>51,7</u>	7%	51,6	7,1%	52,0	7,9%

References

- [1] M. Schwonberg, et al., "Survey on unsupervised domain adaptation for semantic segmentation for visual perception in automated driving," IEEE Access, 2023
- [2] J. Frey, et al., "Continual adaptation of semantic segmentation using complementary 2d-3d data representations," IEEE RAL, 2022
- [3] N. Ravi, et al., "Sam 2: Segment anything in images and videos," 2024.
- [4] A. Dai, et al., "ScanNet: Richly-annotated 3d reconstructions of indoor scenes," in Proc. of CVPR, 2017.